

Mean, Median, Mode & Range

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS For the data set 3, 5, 5, 8, 9: order the numbers first. Mean = add then divide by how many = $(3+5+5+8+9) \div 5 = 6$. Median = middle value when ordered = 5. Mode = value that appears most often = 5. Range = highest – lowest = $9 - 3 = 6$.

Round 1 — Retrieve It

1. How do you find the mean of a data set?

2. How do you find the median of a data set?

3. What does the mode tell you?

4. How do you find the range, and why do you order the numbers first?

Round 2 — Sketch It

Write the data set 3, 5, 5, 8, 9 in order on a number line. Circle the median, and mark the highest and lowest values you would subtract to find the range.

Mean, Median, Mode & Range — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. Use the data set 3, 5, 5, 8, 9. Find the mean. Show the sum and the division.

2. Order 3, 5, 5, 8, 9 and find the median and the mode.

3. Find the range of 3, 5, 5, 8, 9. Show the subtraction.

Round 4 — Reason from Evidence

For the data set 3, 5, 5, 8, 9 the mean and the range are both 6. Explain why one 6 describes a typical value while the other 6 describes how spread out the data is.

Mean, Median, Mode & Range — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the difference between the mean and the median in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect the range to a real situation, such as daily temperatures. Why might a large range matter even when the mean stays the same?

E**Extend**

Create your own data set of five numbers whose mode is 5. Show that you can find its mean, median, mode, and range.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Mean, Median, Mode & Range

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Add all the values, then divide by how many values there are.
2. Order the values, then take the middle value.
3. The mode is the value that appears most often.
4. Range = highest – lowest; ordering the numbers makes it easy to see the highest and lowest.

Sketch It — look for

Numbers placed in order 3, 5, 5, 8, 9. The median 5 circled, with 9 (highest) and 3 (lowest) marked as the values subtracted for the range.

Apply It

1. Mean = $(3+5+5+8+9) \div 5 = 30 \div 5 = 6$.
2. Ordered 3, 5, 5, 8, 9: median = **5**; mode = **5**.
3. Range = $9 - 3 = 6$.

Reason from Evidence — look for

The mean of 6 is a measure of center: it is a single typical value that balances the data. The range of 6 is a measure of spread: it shows the distance from lowest (3) to highest (9). Same number, different meaning.

ACE — Extend look-for

Many answers possible. Check that the data set has five numbers with 5 appearing most often, and that the mean, median, mode, and range are computed correctly.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.